

Matrix Analysis: Rank and Nullity

This study note breaks down how to find the **Rank** and **Nullity** of a matrix using two standard engineering methods: the **Determinant Method** and the **Transformation Method**.

1. Conceptual Breakdown

- **Rank** $\rho(A)$: The number of linearly independent rows. It tells you the "useful" size of the matrix.
 - **Nullity** $N(A)$: The "wasted" dimensions. It represents how many rows became zero during simplification.
 - **The Relation**: $N(A) = \text{Order of Matrix} - \rho(A)$
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2. Method 1: The Determinant Method (Minor Method)

This method involves checking the determinants of the square sub-matrices (minors).

Step 1: Check the 3×3 Determinant

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{vmatrix} = 1(20 - 12) - 2(5 - 4) + 3(6 - 8) = 1(8) - 2(1) + 3(-2) = 8 - 2 - 6 = 0$$

Because $|A| = 0$, the rank is less than 3.

Step 2: Check a 2×2 Minor

Pick any 2×2 sub-matrix, for example, the top left:

$$\begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = (4 - 2) = 2$$

Since we found a minor that is **not zero**, the Rank is **2**.

3. Method 2: Transformation Method (Row Echelon Form)

We use **Elementary Row Operations** to create a staircase of zeros.

The Transformation Algorithm

Step 1: Use R_1 to clear the first column.

We apply $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - 2R_1$ simultaneously:

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix} \xrightarrow{R_2 - R_1, R_3 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix}$$

Step 2: Use R_2 to clear the second column.

Apply $R_3 \rightarrow R_3 - R_2$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

4. Final Calculation

From the final matrix (Echelon Form):

1. **Count Non-Zero Rows:** There are 2 non-zero rows.

- Rank $\rho(A) = 2$

2. **Calculate Nullity:** The matrix order is 3.

- Nullity $N(A) = 3 - 2 = 1$

Quick Summary: Both methods confirm that one row is a combination of the others, leading to a Rank of 2 and a Nullity of 1.